

EXPERIMENT 12

12 To Show that Carbon Dioxide is Given Out During Respiration

Aim

To demonstrate that carbon dioxide gas is released during the process of respiration in germinating seeds.

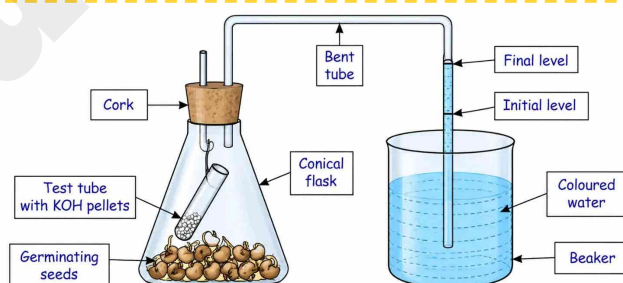
Requirements

A conical flask, beaker, test tube, KOH pellets, a tube bent at right angles, one hole rubber stopper, thread, soaked gram seeds, coloured water, cotton wool, etc.

Procedure

1. Soak gram seeds overnight in water and germinate them by placing the seeds in moist cotton wool for 1 to 2 days.
2. Place moist germinated seeds in conical flask so that respiration may continue for longer time.
3. Suspend a test tube with KOH pellets in the conical flask.
4. Fit the cork with bent glass tube to the conical flask. Keep the connection air tight.
5. Dip the free end of the bent glass tube in a beaker containing coloured water. Mark the initial level of water in the bent tube.
6. Keep this up undisturbed for about 15 minutes and mark the final level of water in the bent tube.
7. Note down the observations.

Diagram



Experiment to show that CO_2 is given out during respiration

Set-up to demonstrate release of carbon dioxide during respiration

Observation

Water in the bent tube rises / ~~falls~~ down. (Underline the correct option and justify your answer.)

The gas evolved during respiration of germinating seeds is absorbed by KOH pellets. This creates a partial vacuum in the conical flask due to which the water level in the bent tube rises.

Inference

Carbon dioxide gas is given out during respiration in plants.

Multiple Choice Questions

Choose the correct alternative and write its letter (A), (B), (C), (D) in the box given:

1. Arrange the steps in correct order to demonstrate that carbon dioxide is produced during respiration.

- (i) Soak gram seeds overnight and place them in a conical flask.
 - (ii) Fit a cork and U tube into the conical flask.
 - (iii) Suspend a test tube containing KOH pellets in the conical flask.
 - (iv) Keep the set up undisturbed for some time.
- (A) i-ii-iii-iv (B) i-ii-iv-iii
(C) i-iii-ii-iv (D) i-iii-iv-i

C

2. Fermentation is a type of:

- (A) Aerobic respiration (B) Anaerobic respiration (C) endothermic respiration (D) None of these

B

3. Before setting up the experiment to show that seeds release CO₂, during respiration, the seeds should be...

- (A) dried Completely (B) Boiled to make them soft (C) soaked in vinegar (D) kept moist till they germinate

D

4. The use of KOH pellets in the experiment is for...

- (A) providing O, required by the seeds for respiration
(B) absorbing CO, and creating partial vacuum in the flask
(C) absorbing water from the seeds to make them dry
(D) making the air present in the flask alkaline

B

5. The process of release of energy from the nutrients is called...

- (A) Respiration (B) Photosynthesis (C) Nutrition (D) Absorption

A

6. The cellular respiration takes place in ...

(A) Lysosome (B) Mitochondrion (C) Chlorophyll (D) Ribosome

B

Remember: Respiration in living organisms may occur with oxygen (aerobic) or without it (anaerobic). Germinating seeds respire aerobically, taking in oxygen and releasing carbon dioxide as a by-product, which is why this gas can be trapped and detected using KOH pellets.

Date: _____

Teacher's Signature: _____